Cisco ISE 802.1X Certificate Authentication for Wi-Fi & Wired via Intune

Implementation Solution Guide

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Executive Summary

The risk

Pre-shared Wi-Fi keys and open switch ports are an open door. A shared passphrase can be copied off a whiteboard or extracted from one device; an unattended network jack lets anything that plugs in reach the corporate network. Neither approach establishes **who** or **what** is connecting. As the fleet grows, this is an unmanaged, unauditable attack surface sitting underneath everything else.

The outcome

Every managed device **proves its identity with a certificate before it is allowed onto the network** — on both **Wi-Fi and wired**. Access is granted by **certificate**, not by a password anyone can share. A device without a valid certificate simply cannot connect. This is on-premises **Network Access Control (NAC)** done the strong way, and it is the wired/wireless counterpart to certificate-based VPN.

The approach

The solution combines two proven platforms across a clean boundary:

- **Microsoft Intune** issues the device certificate (SCEP/PKCS, or Microsoft Cloud PKI) and delivers the **Wi-Fi and wired 802.1X network profiles** to every managed device — automatically, at scale.
- **Cisco ISE** acts as the **RADIUS authentication authority**, validating each device's certificate over **EAP-TLS** and admitting or denying it. The switches and wireless controllers enforce the verdict at the port.

The Intune, Entra, and certificate work is documented at **exact, click-by-click detail**. The Cisco ISE and network-infrastructure configuration is addressed at the **architectural level with pointers to Cisco's official documentation** — this solution never fabricates ISE or switch menu paths.

How it is delivered

A disciplined, reversible rollout: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with three explicit pass/fail gates — **certificate present**, **profile applied**, **ISE 802.1X authentication success** — proven on a small pilot group before any fleet-wide expansion, and a documented rollback at every step.

Included: full requirements, implementation, testing, troubleshooting, and best-practice guides, plus **five PowerShell scripts** for readiness discovery, per-device validation, fleet coverage reporting, safe remediation, and pilot-group targeting — all read-only or `-WhatIf` by default.

The bottom line

Every device certificate-authenticated onto Wi-Fi and wired by Cisco ISE — profiles and certificates delivered from Intune, and proven on a pilot before you scale. Self-guided, tested, and repeatable, at a fraction of the cost of a bespoke consulting engagement.

Solution Overview

What this solution delivers

This solution makes **every managed device prove its identity with a certificate** before it is allowed onto your Wi-Fi or a wired switch port. It does that by using **Microsoft Intune** to issue the device certificate and to push the **Wi-Fi and wired 802.1X network profiles**, and by using **Cisco Identity Services Engine (ISE)** as the **RADIUS authentication authority** that checks the certificate and admits (or denies) the device. The authentication method is **EAP-TLS** — the mutual-certificate flavor of 802.1X — which is the strongest, most phishing-resistant network authentication method in common use.

In plain terms: pre-shared Wi-Fi keys and open switch ports are replaced by **certificate-based network access control (NAC)**. A device that lacks a valid certificate simply cannot get on the network.

How 802.1X / EAP-TLS works (the moving parts)

802.1X is a three-party conversation:

1. **Supplicant** — the software on the endpoint (Windows/macOS/iOS/Android native 802.1X client)

that presents the certificate. Intune configures the supplicant through the Wi-Fi and wired network profiles.

1. **Authenticator** — the network device that controls the port: a **wireless LAN controller (WLC)

or access point **for Wi-Fi, or an** Ethernet switch** for wired. It does not decide who gets in; it relays credentials to the RADIUS server and enforces the verdict (open the port / VLAN, or keep it closed).

1. **Authentication server** — **Cisco ISE**, acting as the **RADIUS** server. ISE validates the

presented certificate against a trusted certificate authority (CA), applies its policy, and returns **Access-Accept** or **Access-Reject** to the authenticator.

With **EAP-TLS**, authentication is *mutual and certificate-based*:

- The **device presents a client certificate** (issued by your CA via Intune).
- **ISE presents its own server certificate**, which the device trusts because Intune also delivered

the **trusted root** for ISE's certificate chain.

- Both sides validate each other's certificate chain. No passwords cross the wire. There is no

pre-shared key to steal, phish, or leak.

The same EAP-TLS mechanism secures **both** transports:

- **Wi-Fi** — the endpoint associates to a corporate SSID configured for **WPA2/WPA3-Enterprise**;

the WLC/AP brokers the EAP-TLS exchange to ISE.

- **Wired** — the endpoint plugs into an 802.1X-enabled switch port; the switch brokers the same

EAP-TLS exchange to ISE.

Because the credential is a certificate on the device (not a user password), this works for **user-attended laptops and unattended/headless devices** alike.

The certificate + network-profile model (what Intune delivers)

Intune's job is to put four things on every in-scope device, all named consistently and reused verbatim throughout this solution:

Object (named)	What it is	Role
`Cisco 802.1X – Trusted Root`	Trusted-certificate profile	Installs the root/intermediate CA that signed **ISE's server certificate**, so the device trusts ISE.
`Cisco 802.1X – SCEP Cert`	SCEP (or PKCS) certificate profile	Enrolls the **device client certificate** used as the 802.1X (EAP-TLS) identity.
`Corp Wi-Fi – EAP-TLS`	Wi-Fi device-configuration profile	Configures the corporate SSID for WPA2/WPA3-Enterprise + EAP-TLS and points the supplicant at the SCEP-issued client cert.
`Wired 802.1X – EAP-TLS`	Wired (Ethernet) device-configuration profile	Configures the wired 802.1X supplicant for EAP-TLS using the same client cert.

The certificate profiles and the network profiles are **linked**: the Wi-Fi and wired profiles are told *which* certificate to present (the one from `Cisco 802.1X – SCEP Cert`) and *which* root to trust for the server (`Cisco 802.1X – Trusted Root`). Get the linkage wrong and the supplicant has nothing to present — the number-one cause of "profile applied but auth fails."

> ***SCEP vs PKCS:*** *Both deliver a device client certificate.* ***SCEP*** *uses the Intune Certificate Connector plus a* ***NDES/SCEP*** *endpoint (or* ***Microsoft Cloud PKI****, which provides a managed CA + SCEP service with no on-prem NDES).* ***PKCS*** *uses the connector against an on-prem enterprise CA. This solution is written for* ***SCEP*** *(the common path, and the one Cloud PKI uses) and notes the PKCS equivalent where it differs. Pick one per certificate profile.*

Where Intune delivery meets Cisco switching / wireless

There is a clean division of responsibility, and this solution respects it deliberately:

- **Intune / Entra / certificate side (authored here in exact click-by-click detail):** issuing the client certificate, delivering the trusted root, and building + assigning the Wi-Fi and wired 802.1X profiles to a pilot group, then expanding.
- **Cisco ISE + network-infrastructure side (architectural level + vendor-doc pointers only):** configuring ISE as the RADIUS authority (policy sets, certificate authentication profile, authorization results), and configuring the **WLC/AP** and **switches** to point at ISE as their RADIUS server and to enable 802.1X on the SSID / ports.

> ***Vendor boundary (important):*** *The exact ISE console screens (policy sets, authorization profiles, certificate authentication profile), and the switch/WLC RADIUS and* `dot1x` *configuration,* ***change across versions and are deferred to Cisco's official documentation****. This guide tells you *what**

*must be true on the Cisco side and *how to verify it from the Intune/device side*; it does **not** fabricate ISE or IOS/AireOS menu paths. Follow Cisco's documentation for the named ISE/switch/WLC step and confirm the precise control in your console.*

The trust chain, end to end

For a device to authenticate successfully, this full chain must line up:

```
Your CA (on-prem AD CS or Microsoft Cloud PKI)
    ■  issues
    ■■■ ISE server certificate  ■■■■■■■■■■■■■■■■ trusted by device via  Cisco 802.1X – Trusted Root
    ■■■ device client certificate  ■■■■■■■■■■■■■ delivered by  Cisco 802.1X – SCEP Cert
                                                    ■ presented over EAP-TLS by
                                          Corp Wi-Fi – EAP-TLS   /   Wired 802.1X – EAP-TLS
                                                    ■ brokered by
                                                WLC / AP   or   802.1X switch  (RADIUS client)
                                                    ■ validated by
                                                  Cisco ISE  (RADIUS authority)
                                                    ■
                                               Access-Accept → device on the network
```

Every downstream document — requirements, implementation, testing, troubleshooting — walks this chain in order and gates each link with a pass/fail check.

Deployment model

This solution follows the standard staged model, with an explicit gate at each step:

Assess → **Configure** → **Pilot** → **Validate** → **Expand.**

- **Assess** — confirm the CA/PKI, Intune certificate connector, ISE RADIUS readiness, and Entra groups (see *Requirements*).
- **Configure** — build the four named Intune objects (trusted root, SCEP cert, Wi-Fi, wired).
- **Pilot** — assign only to `PILOT – 802.1X Devices` and prove the full chain on a handful of devices.
- **Validate** — cert present ✔, profile applied ✔, ISE 802.1X auth success ✔ (the three gates).
- **Expand** — widen assignment to the broader fleet once the pilot passes, keeping a documented rollback.

Requirements

Confirm every prerequisite below **before** you build anything. Each entry follows the same shape — **Description / Why / Verify / Implement if missing** — so you can walk the list top to bottom and close each gap. The category-level prerequisites (Entra tenant, Intune, admin roles, network fundamentals, pilot group, change control) are covered once in `global-requirements-network.md` — read that first; the items below are the ones **specific to 802.1X EAP-TLS with Cisco ISE**.

***Vendor boundary:** Where a requirement lives on the **Cisco ISE, switch, or WLC** side, it is stated at the architectural level with a pointer to Cisco's documentation. Do not expect fabricated ISE/IOS menu paths here — confirm the exact control in your Cisco console.*

R1. A certificate authority (CA) or Cloud PKI

Description: A certificate authority that can issue **device client certificates** (for the EAP-TLS identity) and whose chain also signs **ISE's server certificate**. This is either an on-prem **Active Directory Certificate Services (AD CS)** enterprise CA, or **Microsoft Cloud PKI** (a managed CA + SCEP service in Intune).

Why: EAP-TLS is entirely certificate-based. Without a CA there is nothing to issue the device identity or to anchor the trust between the device and ISE.

Verify: Confirm you have either (a) an AD CS enterprise CA reachable from the Intune Certificate Connector host, or (b) **Microsoft Cloud PKI** provisioned in Intune (Intune admin center → **Tenant administration** → **Cloud PKI**, a CA is listed and *Active*).

Implement if missing: Stand up an AD CS enterprise CA, **or** provision **Microsoft Cloud PKI** (Intune → **Tenant administration** → **Cloud PKI** → **Create** a root or issuing CA). Cloud PKI is the lower-friction path because it removes the on-prem NDES/SCEP server.

R2. Intune Certificate Connector (for SCEP or PKCS)

Description: The **Intune Certificate Connector** installed and healthy, connecting Intune to your SCEP/NDES endpoint (SCEP) or to your enterprise CA (PKCS). *Not required if you use Microsoft Cloud PKI*, which provides the SCEP service natively.

Why: It is the bridge that lets Intune request and deliver certificates from your CA to devices.

Verify: Intune admin center → **Tenant administration** → **Connectors and tokens** → **Certificate connectors**. The connector is listed with a **green/Active** status and a recent last-check-in time.

Implement if missing: Download and install the Certificate Connector on a supported Windows Server that can reach your CA/NDES, sign in with an Intune admin during setup, and confirm it reports *Active*. (Skip entirely if using Cloud PKI.)

R3. Trusted-root certificate profile source

Description: The **root (and any intermediate) CA certificate that signed ISE's server certificate**, exported as a `.cer` file, ready to import into a Trusted certificate profile named `Cisco 802.1X – Trusted Root`.

Why: During EAP-TLS the device must **trust ISE's server certificate**. If the device does not have ISE's issuing root, it rejects the server and authentication fails — even when the device's own certificate is perfect.

Verify: You have the correct root `.cer` on hand, and it matches the chain of the certificate ISE presents for EAP (ask whoever administers ISE which CA signed the **ISE EAP/RADIUS server certificate**).

Implement if missing: Export the root (and intermediates) from your CA as Base-64/DER `.cer` files. If ISE uses its own internal CA or a public CA for its EAP certificate, obtain **that** chain instead — the root must match whatever ISE presents.

R4. SCEP (or PKCS) certificate profile settings

Description: The certificate-request settings for the device client certificate delivered by `Cisco 802.1X – SCEP Cert` — key size, validity, **Subject** and **Subject Alternative Name (SAN)** format, **Extended Key Usage = Client Authentication**, and the **SCEP URL / certification-authority** reference.

Why: ISE matches the incoming certificate against its policy (e.g. it reads a SAN or CN to identify the device). The certificate must contain **Client Authentication** EKU and the identity fields ISE expects, or ISE will reject or misclassify it.

Verify: Agree with the ISE administrator on the **SAN type/value** (commonly a **DNS name** or **UPN**) and CN that ISE's certificate authentication profile will read, plus **Client Authentication** EKU. Confirm your SCEP URL (or Cloud PKI SCEP endpoint) is known.

Implement if missing: Define these values now (they are entered when building `Cisco 802.1X – SCEP Cert` in *Implementation*). For SCEP you also need a matching **NDES/SCEP certificate template**; for Cloud PKI the issuing CA supplies the SCEP endpoint automatically.

R5. Wi-Fi 802.1X profile settings

Description: The parameters for the Wi-Fi profile `Corp Wi-Fi – EAP-TLS`: the **corporate SSID**, **WPA2 or WPA3-Enterprise** security, **EAP-TLS** as the EAP type, the **client certificate** source (`Cisco 802.1X – SCEP Cert`), and the **trusted server certificate/root** (`Cisco 802.1X – Trusted Root`), plus optional trusted **server-name** matching for ISE.

Why: These settings tell the Wi-Fi supplicant *how* to authenticate and *which* certificate to present, and *which* server (ISE) to trust.

Verify: You know the exact **SSID string** (case-sensitive) as configured on the WLC, and whether it is WPA2 or WPA3-Enterprise.

Implement if missing: Obtain the SSID and security mode from the wireless administrator. The SSID must already exist on the WLC and be configured for 802.1X/RADIUS to ISE (see R7).

R6. Wired 802.1X profile settings

Description: The parameters for the wired profile `Wired 802.1X – EAP-TLS`: **EAP-TLS** as the EAP type, the same **client certificate** source (`Cisco 802.1X – SCEP Cert`) and **trusted root** (`Cisco 802.1X – Trusted Root`), and the authentication mode (e.g. user/machine).

Why: Wired 802.1X uses the OS's **Wired AutoConfig** supplicant, which is **not enabled by default** on Windows. The Intune wired profile both enables the supplicant and configures EAP-TLS.

Verify: Confirm the target OS supports Intune wired 802.1X profiles (Windows 10/11 primarily) and that switch ports are (or will be) 802.1X-enabled (see R7).

Implement if missing: Plan the wired profile settings now; they are entered when building `Wired 802.1X – EAP-TLS` in *Implementation*. Ensure **Wired AutoConfig (dot3svc)** will be running (the profile can enforce this).

R7. Cisco ISE configured as the RADIUS authority for EAP-TLS *(vendor side)*

Description: Cisco ISE reachable as a **RADIUS** server, with a **policy set** that permits **EAP-TLS**, a **certificate authentication profile** that reads the certificate field your SCEP profile populates (R4), and an **authorization result** (VLAN/permit) for authenticated devices — plus the **WLC/AP and switches configured as RADIUS clients/NADs** pointing at ISE, with 802.1X enabled on the SSID (R5) and the target switch ports (R6).

Why: ISE is the authority that actually validates the certificate and admits the device. Without the ISE policy and the network devices trusting ISE, the supplicant has nothing to authenticate to.

Verify: Ask the ISE/network administrator to confirm: EAP-TLS is allowed in the relevant policy set; ISE **trusts the CA** that issued the *device* certificates (the CA chain from R1 is in ISE's trusted certificate store); the WLC and switches are defined as **network access devices** with a shared RADIUS secret; and 802.1X is enabled on the SSID and ports.

Implement if missing: This is **Cisco-side work deferred to Cisco documentation** — configure the ISE policy set, certificate authentication profile, authorization profile, and network-device definitions per **Cisco ISE documentation**, and enable 802.1X/`dot1x` + RADIUS on the WLC and switches per **Cisco WLC/IOS documentation**. Do not proceed to Pilot until ISE trusts your device CA and the network devices point at ISE.

R8. Entra security groups (assignment targets)

Description: Entra ID security groups used to target the profiles — at minimum the pilot group `PILOT – 802.1X Devices`, and later a broader production group (e.g. `Corp – 802.1X Devices`).

Why: Intune profiles are assigned to groups. A dedicated group set gives you staged rollout and a clean rollback (remove the group assignment).

Verify: Entra admin center → **Groups**; `PILOT – 802.1X Devices` exists (create it now if not).

Implement if missing: Create the groups in Entra (**Groups** → **New group** → **Security**), or use the provided `ISE8021X-Set-ScopeTargeting.ps1` script to create and populate the pilot group. Use **device** groups for device-certificate/machine authentication.

R9. A pilot group and change control

Description: A defined pilot cohort (`PILOT – 802.1X Devices`) and an approved **change window + rollback path**, as required for every Network solution (see `global-requirements-network.md`, G6/G7).

Why: 802.1X touches live network access — a misconfiguration can lock devices off Wi-Fi or a switch port. Piloting and a rollback plan make the change safe and reversible.

Verify: The pilot group has a small set of representative devices; a rollback (remove profile assignment / switch port fallback) is written and approved.

Implement if missing: Populate `PILOT – 802.1X Devices` with a handful of test devices, and agree a rollback with the network team (e.g. keep a **fail-open** or MAB fallback on pilot switch ports during validation — see *Best Practices*).

Implementation

Follow these steps **in order**. The order matters: the **trusted root** must exist before the SCEP profile references it, the **SCEP certificate** must exist before the Wi-Fi/wired profiles reference it, and everything is assigned to `PILOT – 802.1X Devices` first — never the whole fleet.

All Intune steps use the **Microsoft Intune admin center** at `https://intune.microsoft.com`. Object names are exact and are **reused verbatim** in *Testing* and *Troubleshooting*.

> ***Vendor boundary:*** *Steps 6–7 (ISE and switch/WLC) are stated at the architectural level and* ***deferred to Cisco documentation****. Do not expect ISE/IOS menu paths here.*

Step 0 — Pre-flight

1. Confirm all *Requirements* are green (CA/Cloud PKI, Certificate Connector or Cloud PKI, ISE RADIUS readiness, Entra pilot group).
1. Confirm the Entra group `PILOT – 802.1X Devices` exists and holds a few representative test devices. If not, create it (Entra → **Groups** → **New group** → **Security**) or run `ISE8021X-Set-ScopeTargeting.ps1`.
1. Have on hand: the **root** `.cer` for ISE's server chain (R3), the agreed **SAN/CN format** (R4), the **SSID string** (R5), and your **SCEP URL** (or Cloud PKI issuing CA).

Gate: Do not continue until every item above is confirmed.

Step 1 — Create the trusted-root profile `Cisco 802.1X – Trusted Root`

1. Go to **Devices** → **Manage devices** → **Configuration** → **Create** → **New policy**.
2. **Platform:** choose your target (e.g. **Windows 10 and later**). **Profile type:** **Templates → Trusted certificate**. Click** Create**.
1. **Name:** `Cisco 802.1X – Trusted Root`. **Description:** "Root CA for Cisco ISE EAP server certificate — 802.1X EAP-TLS." Click **Next**.
1. **Configuration settings:** upload the **root** `.cer` file that signed ISE's server certificate (from R3). For Windows, set **Destination store = Computer certificate store – Root**. Click **Next**.
1. **Assignments: Add groups** → `PILOT – 802.1X Devices`. Click **Next**.
2. **Review + create** → **Create.**

> *If ISE's chain has intermediates, repeat Step 1 for each intermediate as a separate Trusted certificate profile (name them* `Cisco 802.1X – Trusted Root (Intermediate)`*, destination store*

Intermediate.

Gate: Profile shows **Assigned**; on a pilot device the ISE root appears under the machine's **Trusted Root Certification Authorities** store.

Step 2 — Create the SCEP certificate profile `Cisco 802.1X – SCEP Cert`

*(If you use PKCS instead of SCEP, choose **PKCS certificate** as the profile type and reference your CA/template instead of a SCEP URL; the identity settings below are the same.)*

1. **Devices** → **Configuration** → **Create** → **New policy.**
2. **Platform:** same target. **Profile type: Templates** → **SCEP certificate**. Click **Create**.
3. **Name:** `Cisco 802.1X – SCEP Cert`. **Description:** "Device client certificate for 802.1X EAP-TLS to Cisco ISE." **Next**.

1. **Configuration settings:**

• **Certificate type: Device** (machine certificate for device/machine authentication).

• **Subject name format:** e.g. `CN={{DeviceName}}` (match what ISE expects).

• **Subject alternative name:** add the type ISE reads (per R4), e.g. **DNS = `{{DeviceName}}` **or** UPN**.

• **Certificate validity period:** e.g. **1 year**.

• **Key storage provider (KSP): Enroll to TPM if present, otherwise software**.

• **Key usage: Digital signature** and **Key encipherment**.

• **Extended key usage:** ensure **Client Authentication (1.3.6.1.5.5.7.3.2)** is present.

• **Root certificate:** select the trusted-root profile `Cisco 802.1X – Trusted Root`.

• **SCEP server URLs:** enter your SCEP/NDES URL, **or** select your **Cloud PKI** issuing CA's SCEP endpoint. Click **Next**.

1. **Assignments: Add groups** → `PILOT – 802.1X Devices`. **Next**.
2. **Review + create** → **Create.**

Gate: Profile shows **Assigned**; on a pilot device a certificate with **Client Authentication** EKU and the expected Subject/SAN appears in the **machine personal (My)** store, chaining to the ISE root.

Step 3 — Create the Wi-Fi 802.1X profile `Corp Wi-Fi – EAP-TLS`

1. **Devices** → **Configuration** → **Create** → **New policy.**
2. **Platform:** target. **Profile type: Templates** → **Wi-Fi**. Click **Create**.
3. **Name:** `Corp Wi-Fi – EAP-TLS`. **Description:** "Corporate Wi-Fi, WPA2/WPA3-Enterprise, EAP-TLS to Cisco ISE." **Next**.

1. **Configuration settings:**

• **Wi-Fi type: Enterprise**.

• **SSID:** the exact corporate **SSID** (case-sensitive, from R5).

• **Connect automatically** = Yes; **Connect to this network even when not broadcasting** as appropriate.

• **Security type: WPA2-Enterprise** (or **WPA3-Enterprise** if supported end-to-end).

• **EAP type: EAP-TLS**.

• **Client certificate for on-client authentication (identity certificate):** select `Cisco 802.1X – SCEP Cert`.

• **Certificate server names / Root certificate for server validation:** select `Cisco 802.1X – Trusted Root`; where available, add ISE's **server name(s)** for server-name matching. Click **Next**.

1. **Assignments: Add groups** → `PILOT – 802.1X Devices`. **Next**.
2. **Review + create** → **Create.**

Gate: Profile shows **Assigned**; a pilot device sees the corporate SSID provisioned and, if in range, associates using the device certificate.

Step 4 — Create the wired 802.1X profile `Wired 802.1X – EAP-TLS`

1. **Devices** → **Configuration** → **Create** → **New policy.**
2. **Platform: Windows 10 and later**. **Profile type: Templates** → **Wired network**. Click **Create**. *(If "Wired network" is not present as a template, use the Settings Catalog "Wired Network" category; the fields map one-to-one.)*

1. **Name:** `Wired 802.1X – EAP-TLS`. **Description:** "Wired 802.1X, EAP-TLS to Cisco ISE." **Next**.
2. **Configuration settings:**

• **Authentication mode:** e.g. **Machine** (device certificate) — match your ISE policy.

• **EAP type: EAP-TLS**.

• **Client certificate (identity):** select `Cisco 802.1X – SCEP Cert`.

• **Trusted server / root certificate:** select `Cisco 802.1X – Trusted Root`; add ISE server-name matching where available.

• Enable **remember/cache** and **single sign-on** settings per your standard; ensure the profile **starts the Wired AutoConfig (dot3svc) service**. Click **Next**.

1. **Assignments: Add groups** → `PILOT – 802.1X Devices`. **Next**.
2. **Review + create** → **Create.**

Gate: Profile shows **Assigned**; on a pilot device the **Wired AutoConfig** service is running and the wired NIC is set to authenticate with EAP-TLS.

Step 5 — Sync a pilot device and confirm delivery

1. On a pilot device: **Settings** → **Accounts** → **Access work or school** → **Info** → **Sync** (or run `ISE8021X-Validation.ps1` locally).
1. Confirm all four objects landed: trusted root present, client cert present (Client Authentication EKU), Wi-Fi profile present, wired profile present + dot3svc running.
1. In Intune, **Devices** → **Configuration**, open each profile → **Device status** shows the pilot device **Succeeded**.

Gate (Intune side): cert present ✔ and all profiles **Succeeded** on the pilot device.

Step 6 — Confirm the Cisco ISE + network side *(vendor side — deferred to Cisco docs)*

Before an authentication can succeed, the ISE/network side (R7) must be true. Confirm with the ISE/network administrator, per **Cisco ISE / WLC / IOS documentation**:

- ISE **trusts the CA** that issued the device certificate (device CA chain in ISE's trusted store).
- An ISE **policy set** permits **EAP-TLS**, with a **certificate authentication profile** reading the certificate field your SCEP profile populated (R4), and an **authorization result** (permit/VLAN) for authenticated devices.
- The **WLC/AP** and target **switches** are defined in ISE as network access devices with the shared RADIUS secret, and **802.1X is enabled** on the SSID and the target switch ports.

Do **not** fabricate these ISE/switch steps — follow Cisco documentation and confirm each control in the Cisco console.

Gate (Cisco side): ISE is reachable as RADIUS, trusts the device CA, and the network devices point at ISE.

Step 7 — Validate end-to-end authentication (pilot)

1. **Wi-Fi:** connect a pilot device to the corporate SSID; confirm it associates **without** a password prompt (certificate is used).
1. **Wired:** plug a pilot device into an 802.1X-enabled port; confirm the port authorizes.
2. In **ISE** → **Operations** → **RADIUS** → **Live Logs** (Cisco side), confirm an **Access-Accept** with **EAP-TLS** for the pilot device. A **reject** here points back to the trust chain or ISE policy (see *Troubleshooting*).

Gate (the three gates together): cert present ✔, profile applied ✔, **ISE 802.1X auth success** ✔. Do not expand until all three pass on multiple pilot devices.

Step 8 — Expand to the fleet

1. Once the pilot passes on Wi-Fi **and** wired across representative device types, add the broader production group (e.g. `Corp – 802.1X Devices`) to the **Assignments** of all four profiles.
1. Expand in **waves** (add device batches to the production group), watching Intune profile status and ISE Live Logs after each wave.
1. Keep `PILOT – 802.1X Devices` assigned for ongoing testing of changes.

Rollback at any point: remove the profile **assignment** from the affected group (and, if necessary, keep a fail-open/MAB fallback on switch ports during rollout) — see *Best Practices* and `global-requirements-network.md` (G7).

Testing & Validation

Testing is organized around the **three gates** used throughout this solution. A device is not "done" until all three pass, on **both** Wi-Fi and wired, across representative device types.

> ***The three gates: (1) cert present → (2) profile applied → (3) ISE 802.1X auth success.*** *They are ordered — a failure at gate 1 or 2 explains most gate-3 failures, so always test in order.*

Run the validation from two angles: **per-device** (`ISE8021X-Validation.ps1` on a pilot device) and **fleet-wide** (`ISE8021X-Report.ps1` against Graph). The Cisco-side confirmation (ISE Live Logs) is **deferred to the Cisco console** and noted where relevant.

Gate 1 — Certificate present (device identity)

Check	How to test	Pass	Fail
Trusted root installed	Machine **Trusted Root Certification Authorities** store (certlm.msc) or `ISE8021X-Validation.ps1`	ISE's issuing root is present	Root missing → Wi-Fi/wired will reject ISE's server cert
Client cert issued	Machine **Personal (My)** store	A cert with **Client Authentication** EKU, expected Subject/SAN, chaining to the ISE root	No cert, or wrong EKU/SAN
Cert not expired	Inspect validity dates	Valid and not near expiry	Expired/near expiry → renewal not working (see Troubleshooting)

Gate 1 pass condition: trusted root present **and** a valid client cert with Client Authentication EKU and the ISE-expected Subject/SAN.

Gate 2 — Profiles applied (supplicant configured)

Check	How to test	Pass	Fail
`Cisco 802.1X – Trusted Root`	Intune → profile → **Device status**; or `ISE8021X-Validation.ps1`	**Succeeded** on device	Error/pending
`Cisco 802.1X – SCEP Cert`	Intune → profile → **Device status**	**Succeeded**	Error → connector/template/SCEP URL issue
`Corp Wi-Fi – EAP-TLS`	`netsh wlan show profiles` / Intune status	Corporate SSID profile present, EAP-TLS	Missing profile
`Wired 802.1X – EAP-TLS`	`netsh lan show profiles`; **dot3svc** running (`sc query dot3svc`)	Wired profile present, EAP-TLS, Wired AutoConfig running	Missing profile or dot3svc stopped

Gate 2 pass condition: all four Intune objects **Succeeded**, both network profiles present with **EAP-TLS**, and **Wired AutoConfig (dot3svc)** running for wired.

Gate 3 — ISE 802.1X authentication success (end to end)

Check	How to test	Pass	Fail
Wi-Fi association	Connect pilot device to corporate SSID	Associates with **no password prompt** (cert used); has network access	Prompts for password / fails to associate
Wired authorization	Plug pilot device into 802.1X-enabled port	Port authorizes; device gets its VLAN/access	Port stays unauthorized / no network
ISE verdict *(Cisco side)*	**ISE** → **Operations** → **RADIUS** → **Live Logs** (per Cisco docs)	**Access-Accept**, method **EAP-TLS**, for the device	**Access-Reject** — read the ISE failure reason

Gate 3 pass condition: Wi-Fi **and** wired both authenticate, and ISE Live Logs show **Access-Accept / EAP-TLS** for the device.

Pilot exit criteria (all must be true before Expand)

- Gate 1, 2, 3 pass on **multiple** pilot devices.
- Both **Wi-Fi and wired** authenticate via EAP-TLS.
- Representative **device types/OS** covered (e.g. Windows laptop + any other in scope).
- **Certificate renewal** verified (a cert nearing expiry re-enrolls; test by shortening validity on a lab device or confirming the renewal threshold).

- **Rollback** exercised once (remove assignment → device falls back cleanly per plan).
- ISE Live Logs are clean (no unexpected rejects) across the pilot window.

Only when every box is checked do you widen assignment to the production group (Implementation, Step 8).

Fleet coverage (ongoing)

Run `ISE8021X-Report.ps1` to produce a coverage report across the fleet:

- devices with the **client certificate** present,
- devices with `Corp Wi-Fi – EAP-TLS` applied,
- devices with `Wired 802.1X – EAP-TLS` applied,
- gaps (in the target group but missing a cert or profile).

Coverage gaps are the input to `ISE8021X-Remediation.ps1` (which defaults to `-WhatIf`).

Troubleshooting

Work the problem in the order of the **three gates** — most authentication failures are actually a missing certificate (gate 1) or a mis-linked profile (gate 2) surfacing as an ISE reject (gate 3). Object names below are the exact ones created in *Implementation*.

> ***Vendor boundary:*** *ISE Live Logs, ISE policy, and switch/WLC configuration are on the* ***Cisco side*** *and are referenced at the architectural level — read the ISE failure reason and the switch port state per* ***Cisco documentation****; this guide does not fabricate ISE/IOS menu paths.*

A. No certificate on the device (gate 1)

Symptom: `ISE8021X-Validation.ps1` reports no client cert, or the machine **Personal (My)** store has no Client-Authentication cert.

- `Cisco 802.1X – SCEP Cert` **shows an error in Intune (Device status).** Check the **Certificate

Connector **health (Tenant administration → Connectors and tokens) — if it is down, no SCEP request succeeds. For Cloud PKI, confirm the issuing CA is** Active**.

- **Connector healthy but still no cert.** Verify the **SCEP URL** is reachable from the device and

the **NDES/SCEP template** matches the profile (key usage, EKU, validity). A template mismatch causes silent enrollment failure.

- **Root not present, so the chain can't build.** Confirm `Cisco 802.1X – Trusted Root` deployed

first and is in the machine Root store; the SCEP profile references it under **Root certificate**.

- **Wrong device targeting.** Confirm the device is in `PILOT – 802.1X Devices` and the profile

assignment includes that group.

B. Profile not applied (gate 2)

Symptom: cert is present but the Wi-Fi or wired profile is missing / shows an error.

• **Profile shows "Error"/"Conflict" in Intune.** Open **Device status** on the specific profile; a **conflict** usually means another Wi-Fi/wired profile targets the same device with different settings — remove the conflicting one.

• **Wi-Fi profile applied but SSID never appears.** The **SSID string is case-sensitive** and must match the WLC exactly. Re-check `Corp Wi-Fi – EAP-TLS` against the real SSID.

• **Wired profile applied but nothing happens.** Confirm **Wired AutoConfig (dot3svc)** is running (`sc query dot3svc`); the wired profile should start it. If stopped, the supplicant never engages the port.

• **Platform mismatch.** A Windows wired profile won't apply to a non-Windows device — confirm the profile **platform** matches the target OS.

• **Not synced yet.** Force a sync (Company Portal / Settings → Access work or school → Sync) and re-check.

C. ISE rejects the authentication (gate 3)

Symptom: cert + profiles present, but the device won't get on the network; ISE Live Logs show **Access-Reject** *(read the ISE reason — Cisco side)*.

• **"Untrusted certificate" / chain not trusted at ISE.** ISE does not trust the **CA that issued the device certificate**. The device CA chain must be in** ISE's trusted certificate store** — a Cisco-side fix per Cisco docs.

• **Device rejects ISE's server certificate.** The device is missing **ISE's** root — confirm `Cisco 802.1X – Trusted Root` is the correct root for ISE's *server/EAP* certificate (not the device CA), and that server-name matching (if enabled) matches ISE's actual name.

• **Identity/SAN mismatch.** ISE's **certificate authentication profile** reads a field (CN, SAN-DNS, SAN-UPN) that your `Cisco 802.1X – SCEP Cert` does not populate the way ISE expects. Align the **Subject/SAN format** (R4) with what ISE reads — coordinate with the ISE admin.

• **EAP-TLS not permitted / wrong policy set.** ISE's policy set for this SSID/port range doesn't allow EAP-TLS or lacks an authorization result — Cisco-side policy fix per Cisco docs.

• **Device unknown to ISE as a NAD path.** The **WLC/switch isn't defined in ISE** as a network access device, or the **shared RADIUS secret** is wrong — Cisco-side fix.

D. Supplicant issues

• **Windows keeps prompting for a password on the corporate SSID.** The supplicant isn't using

EAP-TLS / the client cert. Confirm `Corp Wi-Fi – EAP-TLS` sets **EAP type = EAP-TLS** and points at `Cisco 802.1X – SCEP Cert`; confirm a matching cert exists in the store it looks in (machine vs user).

• **Machine vs user certificate mismatch.** If ISE expects a **machine** cert but the profile presents a **user** cert (or vice-versa), it rejects. Match the **certificate type** (Device/User) and the profile's **authentication mode** to ISE's policy.

• **Server-name validation failing.** If you enabled trusted **server-name** matching, a mismatch with ISE's real certificate name blocks auth — correct the server name(s) in the profile.

• **TPM / key provider errors.** If **Enroll to TPM if present** is set on a device without a working TPM, enrollment can fail — allow software KSP fallback or fix the TPM.

E. Wired vs wireless differences (why one works and the other doesn't)

• **Wired needs Wired AutoConfig (dot3svc); Wi-Fi needs WLAN AutoConfig (wlansvc).** These are *different* services. Wi-Fi can authenticate while wired does nothing simply because **dot3svc is stopped** — start/enforce it via `Wired 802.1X – EAP-TLS`.

• **Different profiles, different fixes.** `Corp Wi-Fi – EAP-TLS` and `Wired 802.1X – EAP-TLS` are separate objects; a fix to one does not apply to the other. Validate and troubleshoot each independently.

• **The switch port vs the SSID.** Wired auth also depends on **802.1X being enabled on that specific switch port **with the right host mode; Wi-Fi depends on the** SSID **being 802.1X-enabled on the WLC. A device that authenticates on Wi-Fi but not wired often just hit a** non-802.1X or differently-configured switch port** — a Cisco-side check.

• **Order of authentication (machine vs user) can differ** between the wired and Wi-Fi profiles; align both with ISE's expected policy.

F. Fast triage checklist

1. Run `ISE8021X-Validation.ps1` on the device → is the **cert** present? (gate 1)
2. Are **both** profiles **Succeeded** in Intune, and is **dot3svc** running? (gate 2)
3. In **ISE Live Logs** *(Cisco side)*, is it **Accept** or **Reject** — and what is the **reason**? (gate 3)

1. Reject reason points the fix: **untrusted cert** → trust chain (device CA in ISE / ISE root on device); **identity mismatch** → SAN/CN vs ISE cert-auth profile; **method not allowed** → ISE policy set; **unknown NAD** → WLC/switch RADIUS definition.

1. If only **one transport** fails, compare the Wi-Fi vs wired profile and the service (wlansvc vs dot3svc) and the network device (SSID vs switch port).

Best Practices

Operating 802.1X EAP-TLS well is mostly about **the supplicant**, **certificate lifecycle**, and choosing the right **failure behavior**. The practices below keep the deployment reliable and reversible.

Supplicant configuration

- **Prefer device (machine) certificates for unattended access.** A machine certificate authenticates the device before/without a logged-in user, which is essential for kiosks, headless devices, and boot-time network access. Reserve user certificates for scenarios that truly need per-user identity on the network.
- **Always enable server-certificate (server-name) validation.** Configure the Wi-Fi and wired profiles to validate ISE's server certificate **and** match ISE's server name(s). Without this the supplicant will trust *any* server presenting a cert from the trusted root — a rogue-RADIUS risk.
- **Deliver the supplicant config from Intune, never by hand.** Both `Corp Wi-Fi – EAP-TLS` and `Wired 802.1X – EAP-TLS` should be the single source of truth. Manual `netsh`/GUI edits drift and are impossible to audit at fleet scale.
- **Ensure Wired AutoConfig (dot3svc) is running** on every wired-802.1X device — the wired profile should start/enforce it. A stopped dot3svc is the classic "wired profile applied but port never authenticates" cause.
- **Keep the certificate ↔ profile linkage intact.** The Wi-Fi/wired profiles must reference `Cisco 802.1X – SCEP Cert` (identity) and `Cisco 802.1X – Trusted Root` (server trust). If you rebuild a cert profile, re-point the network profiles at it.

Certificate lifecycle & renewal

- **Set a sensible validity + renewal threshold.** Typical: 1-year device certificates with SCEP renewal starting at ~20% remaining lifetime, so devices re-enroll well before expiry. Verify renewal actually happens during the pilot — expired certs are the most common cause of a device silently dropping off the network.
- **Watch the Certificate Connector / Cloud PKI health.** Renewals fail silently if the connector is down or the SCEP endpoint is unreachable. Monitor connector status; alert on it.
- **Plan for CA rotation.** If ISE's issuing CA or your device CA is rotated, deploy the **new root** via a new/updated `Cisco 802.1X – Trusted Root` **before** the old chain is retired, and ensure ISE trusts the new device CA — otherwise a rotation locks the fleet off the network.
- **Protect keys with the TPM.** Prefer **Enroll to TPM if present** so the private key is non-exportable and bound to the device.

Fail-open vs fail-closed (choose deliberately)

This is a network-infrastructure decision (configured on the **switch/WLC**, Cisco-side) but you must choose it consciously:

• **Fail-closed (default-deny):** an unauthenticated device gets **no access**. Most secure; also the most disruptive if certificates or ISE break. Appropriate for high-security segments once the deployment is proven.

• **Fail-open / low-impact modes:** during rollout, network teams often run 802.1X in **monitor mode **(a.k.a. open/low-impact) or with a** MAB (MAC Authentication Bypass) fallback **and/or a** critical-VLAN **so that an ISE outage or a cert problem doesn't strand devices. Use this during** Pilot and early Expand**, then tighten toward fail-closed once confidence is high.

• **Always have a rollback.** Keep the ability to remove the Intune profile assignment *and* a network-side fallback (fail-open/MAB/critical-VLAN) so a bad change never becomes an outage. This satisfies the category rollback requirement (`global-requirements-network.md`, G7).

> ***Vendor boundary:*** *the exact monitor-mode/MAB/critical-VLAN configuration lives on the Cisco switch/WLC and in ISE — configure it per* ***Cisco documentation****.*

Rollout discipline

• **Pilot first, expand in waves.** Assign to `PILOT – 802.1X Devices`, prove all three gates on Wi-Fi and wired, then widen to the production group in batches — never the whole fleet at once.

• **Test both transports and multiple OS types.** Wi-Fi and wired behave differently (different supplicant, dot3svc for wired); validate each.

• **Keep names consistent.** Reuse the exact object names so reports, remediation, and troubleshooting line up across tools and teammates.

Operations & monitoring

• **Run the coverage report regularly.** `ISE8021X-Report.ps1` surfaces devices missing a cert or a profile before users notice. Feed gaps into `ISE8021X-Remediation.ps1` (default `-WhatIf`).

• **Watch ISE Live Logs after every wave and every ISE/CA change** (Cisco-side) — a spike in rejects is the earliest warning of a trust-chain or policy regression.

• **Document the trust chain.** Record which CA issues device certs, which CA signs ISE's server cert, and the SAN/CN format ISE reads — so a future admin can reason about a reject in minutes.

Deliverables Summary

Everything included in this solution, and where to use it.

Documentation

Document	Purpose
Executive Summary	One-page business framing: the risk, the outcome, the approach.
01 – Solution Overview	How 802.1X / EAP-TLS works for Wi-Fi and wired with Cisco ISE as RADIUS authority; the certificate + network-profile model; where Intune delivery meets Cisco switching/wireless.
02 – Requirements	Every 802.1X-specific prerequisite with **Description / Why / Verify / Implement-if-missing** (CA/Cloud PKI, Certificate Connector, trusted root, SCEP/PKCS, Wi-Fi/wired settings, ISE RADIUS readiness, Entra groups, pilot).
03 – Implementation	Exact, ordered Intune steps: trusted root → SCEP cert → Wi-Fi 802.1X → wired 802.1X → assign to pilot → confirm ISE → validate → expand.
05 – Testing & Validation	The three pass/fail gates (cert present, profile applied, ISE auth success) and pilot exit criteria.
06 – Troubleshooting	The common failure modes: no cert, profile not applied, ISE rejects, supplicant issues, wired-vs-wireless differences.
07 – Best Practices	Supplicant, certificate renewal, and fail-open vs fail-closed guidance.
Global Requirements (Network)	Category-level prerequisites shared by all Network solutions.

Named objects created (reused verbatim across all docs and scripts)

Object	Type	Role
`Cisco 802.1X – Trusted Root`	Intune Trusted certificate profile	Installs the root that signed ISE's server certificate.
`Cisco 802.1X – SCEP Cert`	Intune SCEP (or PKCS) certificate profile	Issues the device client certificate (EAP-TLS identity).
`Corp Wi-Fi – EAP-TLS`	Intune Wi-Fi profile	WPA2/WPA3-Enterprise + EAP-TLS corporate SSID.
`Wired 802.1X – EAP-TLS`	Intune Wired network profile	Wired 802.1X EAP-TLS supplicant.
`PILOT – 802.1X Devices`	Entra security group	Pilot assignment target.
`Corp – 802.1X Devices`	Entra security group	Production assignment target (Expand phase).

PowerShell scripts

All scripts are **read-only** against Microsoft Graph except remediation/scope-targeting, which **default to** `-WhatIf`. Cisco ISE and network-infrastructure internals are **deferred to vendor documentation** — no script touches ISE, switches, or the WLC.

Script	What it does
ISE8021X-Discovery.ps1	Tenant-wide 802.1X readiness: certificate connectors, and the trusted-root / SCEP / PKCS / Wi-Fi / wired 802.1X profiles present in Intune.
ISE8021X-Validation.ps1	On **this** device: confirms the client certificate and the Wi-Fi + wired 802.1X profiles are applied (the three gates, device side).
ISE8021X-Report.ps1	Fleet coverage: which devices have the certificate and the Wi-Fi/wired profiles, and where the gaps are.
ISE8021X-Remediation.ps1	Safely fixes missing profile/cert assignments (adds devices/groups to the right profiles). **Default** `-WhatIf`**.**
ISE8021X-Set-ScopeTargeting.ps1	Creates/populates the pilot Entra group(s) used as assignment targets. **Default** `-WhatIf`**.**

Deployment model

Assess → **Configure** → **Pilot** → **Validate** → **Expand**, gated by **cert present** → **profile applied** → **ISE 802.1X auth success**, with a documented rollback (remove assignment + network-side fallback) at every step.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

ISE8021X-Discovery.ps1

Purpose: Read-only discovery of whether the tenant is ready for Cisco ISE 802.1X EAP-TLS. Reports the Intune certificate connectors and the device-configuration profiles that matter: trusted-root, SCEP/PKCS certificate, Wi-Fi (EAP-TLS) and wired 802.1X. Nothing is changed. Cisco ISE / switch / WLC internals are out of scope (vendor docs). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ISE8021X-Discovery.ps1

How to run: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Discovery.ps1

ISE8021X-Remediation.ps1

Purpose: Closes coverage gaps found by ISE8021X-Report.ps1 by ensuring the trusted-root, SCEP cert, Wi-Fi EAP-TLS and wired 802.1X profiles are all assigned to the intended target group. It only adds a group assignment where one is missing; it never deletes, never edits profile settings, and never touches Cisco ISE/switch/WLC. Runs in -WhatIf mode by default — pass -Apply to make changes. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\ISE8021X-Remediation.ps1 -TargetGroup 'PILOT - 802.1X Devices' Apply: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Remediation.ps1 -TargetGroup 'PILOT - 802.1X Devices' -Apply

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\ISE8021X-Remediation.ps1 -TargetGroup 'PILOT - 802.1X Devices' Apply: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Remediation.ps1 -TargetGroup 'PILOT - 802.1X Devices' -Apply

ISE8021X-Report.ps1

Purpose: Read-only reporting of 802.1X EAP-TLS coverage across the managed fleet. For each relevant Intune profile (trusted-root, SCEP/PKCS cert, Wi-Fi EAP-TLS, wired 802.1X) it reports how many devices report Succeeded, and highlights devices assigned but not yet compliant. Nothing is changed. Cisco ISE authentication results live in the ISE console (vendor docs). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ISE8021X-Report.ps1 -OutputCsv .\ise8021x-coverage.csv

How to run: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Report.ps1 -OutputCsv .\ise8021x-coverage.csv

ISE8021X-Set-ScopeTargeting.ps1

Purpose: Creates the pilot Entra security group (default 'PILOT - 802.1X Devices') if it does not exist, and optionally adds named devices to it, so the 802.1X trusted-root, SCEP cert, Wi-Fi and wired profiles have a clean assignment target. Runs in -WhatIf mode by default — pass -Apply to make changes. Does not touch Cisco ISE/switch/WLC. HOW TO RUN Preview: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Set-ScopeTargeting.ps1 -DeviceNames 'LT-0001','LT-0002' Apply: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Set-ScopeTargeting.ps1 -DeviceNames 'LT-0001','LT-0002' -Apply

How to run: Preview: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Set-ScopeTargeting.ps1 -DeviceNames 'LT-0001','LT-0002' Apply: powershell -ExecutionPolicy Bypass -File

.\ISE8021X-Set-ScopeTargeting.ps1 -DeviceNames 'LT-0001','LT-0002' -Apply

ISE8021X-Validation.ps1

Purpose: Read-only, local validation of the three device-side gates for Cisco ISE 802.1X EAP-TLS: (1) client certificate present with Client Authentication EKU, (2) Wi-Fi and wired 802.1X profiles applied with EAP, (3) Wired AutoConfig (dot3svc) running. Runs on the device; no Graph connection required. ISE-side verdict is checked in the ISE console (vendor docs). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\ISE8021X-Validation.ps1

How to run: powershell -ExecutionPolicy Bypass -File .\ISE8021X-Validation.ps1